Lettuce_growth_days

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Contents

Install packages and attach libraries

```
# install.packages("psych")
# install.packages("lubridate")
# install.packages("purrr")
# install.packages("GGally")
# install.packages("skimr")
# install.packages("Writexl")
library(psych)
library(googlesheets4)
library(tidyr)
library(dplyr)
library(janitor)
library(ggplot2)
library(lubridate)
library(purrr)
library(GGally)
library(readxl)
library(skimr)
library(writexl)
```

Step 1. Ask

1.1 Key tasks and Deliverable(s)

1.1.1 A clear statement of the business task

Determine how environmental factors (temperature, humidity, and total dissolved solids) influence lettuce growth and maturity time, in order to optimize greenhouse conditions and predict growth outcomes for future growing seasons.

1.1.2 Key stakeholders

Managers

1.1.3 Smart questions

N	Type	Questions
1	Specific	Is there a connection between humidity and temperature?
2	Measurable	How much growth is normal or expected over a one week period?

N	Type	Questions
3	Action-oriented	If total dissolved solids are increased when growing lettuce, will it shorten the time needed for the plants to reach maturity?
4	Relevant	Based on the dataset, how can predictions about growth be made given the conditions?
5	Time - bound	1) What is the data range for this data set? 2) Do we need an understanding of this data for next season?

1.1.4 Scope of work (SOW)

N	Deliverables	Timeline	Milestones	Results	What SMART questions will be answered
1	Clean dataset	Day 1	Completion of data download and preparation	Dataset is clean and usable	-
2	Exploratory data analysis (EDA) findings (form: spreadsheet, google doc summary)	Day 2	Completion of EDA - understanding of dataset is formed	EDA highlights room for investigation. Questions were formulated based on the results of EDA.	**Specific:** Is the connection between humidity and temperature **Time - Bound:** What is the data range for this data set?
3	Engineered features in the form of new columns (examples may include Avg 3 day growth, Growth since day before, etc.)	Day 3	Completion of feature engineering	New columns engineered to provide more accessible information	**Measurable:** How much growth is normal or expected over a one week period?

N	Deliverables	Timeline	Milestones	Results	What SMART questions will be answered
4	Statistical insights	Day 3	Completion of statistical analysis	Statistical analysis conducted	**Action-oriented:** If total dissolved solids are increased when growing lettuce, will it shorten the time needed for the plants to reach maturity? **Relevant:** Based on the dataset, how can predictions about growth be made given the conditions? **Time - bound:** Do we need an understanding of this data for next season?
5	Visual Aids (images, dashboards)	Days 4-5	Completion of visualizations	Visualizations created	-
6	Final report (spreadsheet and PowerPoint)	Day 6-10	Final report submission	Final report completed	-

1.2 Guiding questions

N	Questions	Answers
1	What is the problem you are trying to solve?	The challenge lies in understanding (finding patterns) how environmental conditions influence lettuce growth time. This knowledge will help inform decisions to optimize growth cycle.
2	How can you insights drive business decisions?	Knowing about environmental conditions that impact lettuce growth can help managers decide on construction of an appropriate greenhouse

Step 2. Prepare

2.1 Key tasks and deliverable(s)

2.1.1 Download data and store it appropriately

Data downloaded and stored on my personal computer and Google Drive. Data available at these links: 1) On the work computer: C:/Users/GudievZK/Desktop/GitHub/Google Data Analytics Professional Certificate/1. Ask_Questions_to_Make_Data_Driven_Decisions/lettuce_dataset_SOW.xlsx, sheet =

“source_lettuce_dataset 2) On the laptop: /Users/zelimkhan/Desktop/Data/GitHub/Google Data Analytics Professional Certificate/1. Ask_Questions_to_Make_Data_Driven_Decisions/lettuce_dataset_SOW.xlsx 3) On the Google Drive

2.1.2 Identify how it’s orginized

The data is organized as a table containing data on 70 plants. The rows contain environmental conditions for each of the 70 plants on each day throughout the growing season. The columns contain information about the plant identifier (Plant_ID) and the following environmental conditions, which were recorded daily for each of the 70 plants throughout the growing season: Date; Temperature (?C); Humidity (%); TDS Value (ppm); pH Level; Growth Days.

2.1.3 Sort and filter the data

Data filtered and sorted

2.1.4 Determine the credibility of the data

The following methodologies/frameworks were used to determine the credibility of the data: ROCCC; Data Integrity.

2.1.4.1 ROCCC

N	ROCCC Dimension	Description	Conclusion	Comments
1	Reliable	Source is dependable and consistent.	Watch conclusion below the table	It depends on ROCCC dimensions listed below
2	Original	We get data from original source (or If we obtained data from a secondary or tertiary source, we could compare and verify it with the primary source)	The data is not original	We do not receive data from the original source, we also cannot identify the original source and have no way to compare it with the original data. However, since the data is provided and used for educational purposes, we may use it for educational purposes

N	ROCCC Dimension	Description	Conclusion	Comments
3	Comprehensive	Covers all necessary cases and variables.	It is not possible to determine the level of comprehensiveness of the data	It is logical to assume that data is greatly simplified for educational purposes and does not contain all necessary information needed to solve a business problem. Since it includes the following key variables it can be used for our educational purposes: Plant_ID; Date; Temperature (°C); Humidity (%); TDS Value (ppm); pH Level; Growth Days.
4	Current	The data sources aren't out of date or irrelevant	The data is current	The data can be considered current, even though the observation was conducted in 2023 and the analysis using this information is being carried out in 2025.
5	Cited	Data source is cited and vetted	The source of the data was not cited or verified.	However, since the data is provided and used for training purposes the data can be used in our work.

Reliable: The data isn't reliable

Conclusion on the credibility of the data in accordance with the ROCCC framework: Data isn't credible.

2.1.4.2 Data integrity

N	Data integrity Dimension	Description	Conclusion	Comments
1	Accuracy	The degree to which the data conforms to the actual entity being measured or described	It is not possible to determine the data's accuracy.	Since the data does not contain unrealistic values (such as days with negative growth, numerical values falling within an unexpected data range, or other impossible measurements), the data can be considered accurate.
2	Completeness	Data completeness is a dimension of data quality that measures whether all necessary or required data is present and available for a dataset to fulfill its intended purpose.	It is not possible to determine the level of Completeness of the data	It is logical to assume that data is greatly simplified for educational purposes and does not contain all necessary information needed to solve a business problem. Since it includes the following key variables and doesn't have missing values it can be used for our educational purposes: Plant_ID; Date; Temperature (°C); Humidity (%); TDS Value (ppm); pH Level; Growth Days.
3	Consistency	Data follows the same format and rules.	Data is consistency	The Date column has character type.
4	Trustworthiness	Data trustworthiness refers to the degree to which information can be relied upon as accurate, credible, and secure, allowing users to confidently make decisions and conduct analyses based on it.	The data isn't trustworthiness	The data isn't trustworthy due to low scores for data integrity dimensions listed above.

Trustworthiness: The data isn't trustworthiness.

Conclusion on credibility of the data in accordance with the Data integrity conception: The data isn't credible.

2.1.4.3 Bias The dataset appears to come from a controlled educational or research environment, so it may not present all real world farming conditions or varieties of lettuce. However, since it is used for analytical practice rather than policy or production decisions, the risk of bias affecting conclusions is minimal.

Conclusion on the admissibility of using data: The data are acceptable for use despite negative findings regarding the credibility of the data in accordance with both the ROCCC framework and the concept of data integrity, as well as the inability to demonstrate the absence of bias, as they are used for educational purposes.

2.1.5 A description of all data sourced used

For this analytical task, the “lettuce growth” dataset from Kaggle was used.

2.2 Guiding questions

N	Questions	Answers
1	Where is your data located?	The data is publicly available in Kaggle website at this link
2	How is the data organized?	The data is organized as a table containing data on 70 plants in long format. The rows contain environmental conditions for each of the 70 plants on each day throughout the growing season. The columns contain information about the plant identifier (Plant_ID) and the following environmental conditions, which were recorded daily for each of the 70 plants throughout the growing season: Date; Temperature (?C); Humidity (%); TDS Value (ppm); pH Level; Growth Days.
3	Are there issues with bias or credibility in this data? Does your data ROCCC?	See section “2.1.4 Determine the credibility of the data”

N	Questions	Answers
4	How are you addressing licensing, privacy, security, and accessibility?	1. **Licensing** The lettuce dataset is published under the Apache 2.0.It means: 1) You are allowed to freely use, modify, and share the dataset, including for commercial and non-commercial purposes. 2) You must provide proper attribution to the original authors when you use the data. 3) If you modify the dataset and distribute your modified version, you must indicate what changes you made. 4) There is no warranty—the authors are not responsible for any issues from using the data. 4) The dataset can be combined with other work, even proprietary. However, on the Kaggle page for this dataset, there is an additional statement: "Do not use it for commercial purposes." This suggests the author is placing an extra restriction that is stricter than the typical Apache 2.0 terms. In this case, you should follow the conditions stated on the dataset's page. For academic or personal projects, with attribution, you are free to use the data. 2. **Privacy** It does not contain personally identifiable information (PII), so privacy is not a concern. 3. **Security** Security risks are minimal, as the dataset stored locally on my Google Drive. 4. **Accessibility** Accessibility is ensured by keeping the data in standard formats (csv, xlsx, R data frames), which can be opened in widely available tools such as Excel, Rstudio or Google Sheets.

N	Questions	Answers
5	How did you verify the data's integrity?	Data integrity involves the accuracy, completeness, consistency, and trustworthiness of data throughout its lifecycle. 1. **Accuracy:** I inspected the dataset for unrealistic values (e.g., negative growth days, numerical values falling within an unexpected data range or impossible measurements) using: Sorting and filtering in spreadsheet; R functions such as summary() and is.na(). *Definition Data accuracy refers to the degree to which data correctly represents the true, real-world information or events it describes, meaning it is correct, precise, and free from errors or distortions. Accurate data serves as a trustworthy foundation for effective decision-making, reliable analysis, and efficient operations, making it a critical component of overall data quality.* 2. **Completeness:** I checked for missing or empty cells using: Sorting and filtering in spreadsheet; R functions such as summary() and is.na(). *Definition Data completeness is a dimension of data quality that measures whether all necessary or required data is present and available for a dataset to fulfill its intended purpose. It ensures there are no missing values or gaps in critical information, preventing skewed analyses and enabling accurate decision-making.* 3. **Consistency:** I reviewed column names, formats, and data types to ensure that values were stored uniformly (e.g., all numeric columns have numeric data types). *Definition Data consistency - is the quality of data follows the same format and rules remaining accurate, uniform, and coherent across all systems, databases, and applications within an organization.*. 4. **Trustworthiness:** Users can depend on the data to be correct and dependable. Since the dataset is a clean educational sample, I verified its source and confirmed it had not been altered during download or import (no unexpected changes in row count or structure). *Definition Data trustworthiness refers to the degree to which information can be relied upon as accurate, credible, and secure, allowing users to confidently make decisions and conduct analyses based on it.* Data?s integrity will be checked in more detail (accuracy, completeness, consistency, trustworthiness) in the next step of analyses (step 3 ?Process?).

N	Questions	Answers
6	How does it help you answer you questions?	The dataset directly contains the variables needed (growth days and environmental factors), so it enables me to explore how growth varies by environmental conditions). It provides the empirical basis for descriptive statistics and visualizations that can answer the practice assignment's research questions.
7	Are there any problems with the data?	The Date column has character type instead date.

Step 3. Process (EDA)

Stage 1 of EDA: Data Discovery & Profiling

```
df_0 <- read_sheet("https://docs.google.com/spreadsheets/d/1esw9BCXZD01h4hwuaKowia5eVmflEKTS6NiouamSYA0/

# df_0 <- read_excel("C:/Users/GudievZK/Desktop/GitHub/Google Data Analytics Professional Certificate/1
```

Load data

```
df_0
```

Table overview

```
## # A tibble: 3,169 x 7
##    Plant_ID Date  `Temperature (°C)` `Humidity (%)` `TDS Value (ppm)` `pH Level`
##       <dbl> <chr>              <dbl>          <dbl>             <dbl>      <dbl>
##  1        1 9/12~               31.2             62               576        6.7
##  2        2 9/12~               31.2             66               657        6.8
##  3        3 9/12~               31.2             69               403        6.6
##  4        4 9/12~               31.2             72               729        6.6
##  5        5 9/12~               31.2             72               738        6.2
##  6        6 9/12~               31.2             57               793        6.4
##  7        7 9/12~               31.2             52               468        6.7
##  8        8 9/12~               31.2             51               768        6.1
##  9        9 9/12~               31.2             80               793        6.6
## 10       10 9/12~               31.2             58               794        6.6
## # i 3,159 more rows
## # i 1 more variable: `Growth Days` <dbl>
```

```
colnames(df_0)
```

Column names

```
## [1] "Plant_ID"          "Date"              "Temperature (°C)" "Humidity (%)"
## [5] "TDS Value (ppm)"   "pH Level"          "Growth Days"
```

```
dim(df_0)
```

Number of rows and columns

```
## [1] 3169    7
```

```r
glimpse(df_0)
```

Data types

```
## Rows: 3,169
## Columns: 7
## $ Plant_ID           <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, ~
## $ Date               <chr> "9/12/2023", "9/12/2023", "9/12/2023", "9/12/2023",~
## $ `Temperature (°C)` <dbl> 31.2, 31.2, 31.2, 31.2, 31.2, 31.2, 31.2, 31.2, 31.~
## $ `Humidity (%)`     <dbl> 62, 66, 69, 72, 72, 57, 52, 51, 80, 58, 66, 64, 51,~
## $ `TDS Value (ppm)`  <dbl> 576, 657, 403, 729, 738, 793, 468, 768, 793, 794, 4~
## $ `pH Level`         <dbl> 6.7, 6.8, 6.6, 6.6, 6.2, 6.4, 6.7, 6.1, 6.6, 6.6, 6~
## $ `Growth Days`      <dbl> 41, 41, 41, 41, 41, 41, 41, 41, 41, 41, 41, 41, 41,~
```

```r
sapply(df_0, class)
```

Data classes

```
##         Plant_ID             Date Temperature (°C)     Humidity (%)
##        "numeric"      "character"        "numeric"        "numeric"
##  TDS Value (ppm)         pH Level      Growth Days
##        "numeric"        "numeric"        "numeric"
```

Checking completeness

Number of missing values in each column

```r
colSums(is.na(df_0))
```

```
##         Plant_ID             Date Temperature (°C)     Humidity (%)
##                0                0                0                0
##  TDS Value (ppm)         pH Level      Growth Days
##                0                0                0
```

Checking duplicates

Number of duplicate values in each column

```r
sapply(df_0, function(x) sum(duplicated(x)))
```

```
##         Plant_ID             Date Temperature (°C)     Humidity (%)
##             3099             3121             3071             3138
##  TDS Value (ppm)         pH Level      Growth Days
##             2768             3160             3121
```

Checking uniqueness

Number of unique rows

```r
distinct(df_0) %>% dim()
```

```
## [1] 3169    7
```

Number of unique values in each column

```r
sapply(df_0, function(x) sum(n_distinct(x)))
```

```
##         Plant_ID             Date Temperature (°C)     Humidity (%)
##               70               48               98               31
```

```
##  TDS Value (ppm)         pH Level      Growth Days
##              401                9               48
```

Transorm data Convert Date column from character into date

```
df_0$Date <- mdy(df_0$Date)
```

Convert Plant_ID column from numeric into factor

```
df_0$Plant_ID <- as.factor(df_0$Plant_ID)
```

Clean names

```
df <- clean_names(df_0)
```

Checking data after transormation Table overview

```
df %>% arrange(date)
```

```
## # A tibble: 3,169 x 7
##    plant_id date       temperature_c humidity_percent tds_value_ppm p_h_level
##    <fct>    <date>             <dbl>            <dbl>         <dbl>     <dbl>
##  1 1        2023-08-03          33.4               53           582       6.4
##  2 2        2023-08-03          33.4               74           586       6.1
##  3 3        2023-08-03          33.4               51           667       6.2
##  4 4        2023-08-03          33.4               62           547       6.7
##  5 5        2023-08-03          33.4               67           651       6.3
##  6 6        2023-08-03          33.4               66           795       6.1
##  7 7        2023-08-03          33.4               74           459       6.5
##  8 8        2023-08-03          33.4               76           650       6
##  9 9        2023-08-03          33.4               72           693       6.4
## 10 10       2023-08-03          33.4               66           503       6.5
## # i 3,159 more rows
## # i 1 more variable: growth_days <dbl>
```

Checking completeness after transormation

```
colSums(is.na(df))
```

```
##         plant_id             date    temperature_c humidity_percent
##                0                0                0                0
##    tds_value_ppm        p_h_level      growth_days
##                0                0                0
```

Checking data stricture

```
str(df)
```

```
## tibble [3,169 x 7] (S3: tbl_df/tbl/data.frame)
##  $ plant_id        : Factor w/ 70 levels "1","2","3","4",..: 1 2 3 4 5 6 7 8 9 10 ...
##  $ date            : Date[1:3169], format: "2023-09-12" "2023-09-12" ...
##  $ temperature_c   : num [1:3169] 31.2 31.2 31.2 31.2 31.2 31.2 31.2 31.2 31.2 31.2 ...
##  $ humidity_percent: num [1:3169] 62 66 69 72 72 57 52 51 80 58 ...
##  $ tds_value_ppm   : num [1:3169] 576 657 403 729 738 793 468 768 793 794 ...
##  $ p_h_level       : num [1:3169] 6.7 6.8 6.6 6.6 6.2 6.4 6.7 6.1 6.6 6.6 ...
##  $ growth_days     : num [1:3169] 41 41 41 41 41 41 41 41 41 41 ...
```

Interpretation of the 1st stage EDA: Data Discovery & Profiling **Purpose and Approach**

This stage focused on initial understanding and validation of the lettuce dataset to answer two main questions:

What does the data contain?

Is the data clean and ready for analysis?

Findings

Dataset structure: The table contained 3,169 rows and 7 columns, representing multiple measurements for 70 individual plants across 48 observation days. Columns included: plant_id, date, temperature_c, humidity_percent, tds_value_ppm, p_h_level and growth_days

Uniqueness and duplication checks:

No duplicated rows were found (distinct(df_0) returned 3,169 unique rows).

Each variable had a meaningful count of unique values (e.g., 70 plant IDs, 98 temperature readings, 9 pH levels), confirming valid variability across measurements.

Data type consistency:

The "date" column was initially stored as a character type and was converted to a proper Date format using mdy().

The "plant_id" column was converted from numeric to factor, which correctly represents categorical identifiers.

Column names were standardized with clean_names() for consistent syntax.

Quality assessment:

No missing values were reported.

The only detected issue before transformation was inconsistent date formatting.

After cleaning, all variables were in the correct type and ready for deeper analysis.

Concusion

The dataset was confirmed to be accurate, consistent, and complete. Minor formatting issues (dates, variable types) were corrected.

Step 4. Analyze

Stage 2 of EDA: Univariate analysis ("Understanding individual variables" phase)

Summary statistics Summary statistics using skim function

```
skim(df) %>%
  mutate(across(where(is.numeric), ~ round(.x, 2)))
```

Table 7: Data summary

Name	df
Number of rows	3169
Number of columns	7
Column type frequency:	
Date	1
factor	1
numeric	5

Group variables	None

Variable type: Date

skim_variable	n_missing	complete_rate	min	max	median	n_unique
date	0	1	2023-08-03	2023-09-19	2023-08-25	48

Variable type: factor

skim_variable	n_missing	complete_rate	ordered	n_unique	top_counts
plant_id	0	1	FALSE	70	4: 48, 55: 48, 3: 47, 7: 47

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
temperature_c	0	1	28.14	4.67	18	23.6	30.2	31.5	33.5	
humidity_percent	0	1	64.87	8.99	50	57.0	65.0	73.0	80.0	
tds_value_ppm	0	1	598.05	115.71	400	498.0	593.0	699.0	800.0	
p_h_level	0	1	6.40	0.23	6	6.2	6.4	6.6	6.8	
growth_days	0	1	23.14	13.08	1	12.0	23.0	34.0	48.0	

Summary statistics using summary function

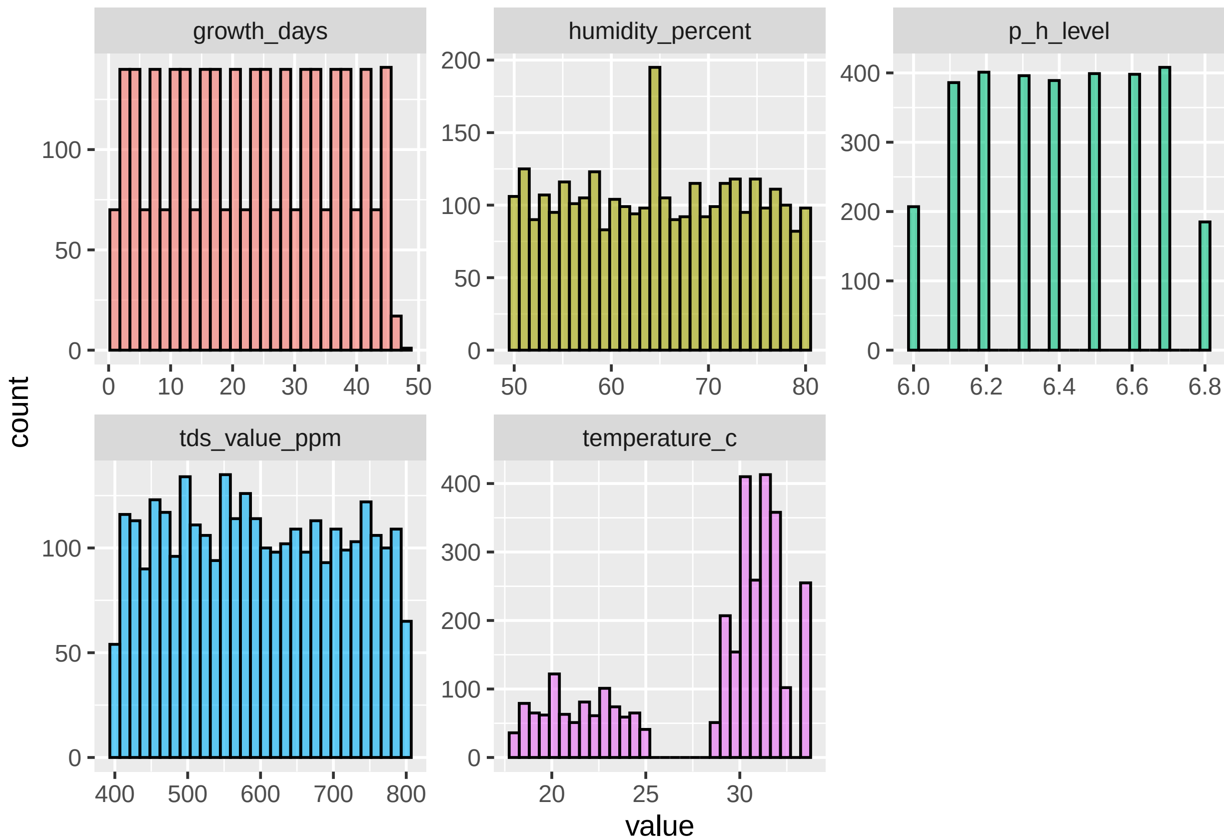
```r
summary(df)
```

```
##     plant_id         date              temperature_c   humidity_percent
##  4      :  48   Min.   :2023-08-03   Min.   :18.00   Min.   :50.00
##  55     :  48   1st Qu.:2023-08-14   1st Qu.:23.60   1st Qu.:57.00
##  3      :  47   Median :2023-08-25   Median :30.20   Median :65.00
##  7      :  47   Mean   :2023-08-25   Mean   :28.14   Mean   :64.87
##  11     :  47   3rd Qu.:2023-09-05   3rd Qu.:31.50   3rd Qu.:73.00
##  48     :  47   Max.   :2023-09-19   Max.   :33.50   Max.   :80.00
##  (Other):2885
##  tds_value_ppm   p_h_level      growth_days
##  Min.   :400   Min.   :6.000   Min.   : 1.00
##  1st Qu.:498   1st Qu.:6.200   1st Qu.:12.00
##  Median :593   Median :6.400   Median :23.00
##  Mean   :598   Mean   :6.399   Mean   :23.14
##  3rd Qu.:699   3rd Qu.:6.600   3rd Qu.:34.00
##  Max.   :800   Max.   :6.800   Max.   :48.00
##
```

Examine distribution and detecting outliers Examine distribution plotting histograms

```r
df %>% select(where(is.numeric)) %>%
  pivot_longer(everything(),
               names_to = "variable",
               values_to = "value") %>%
```

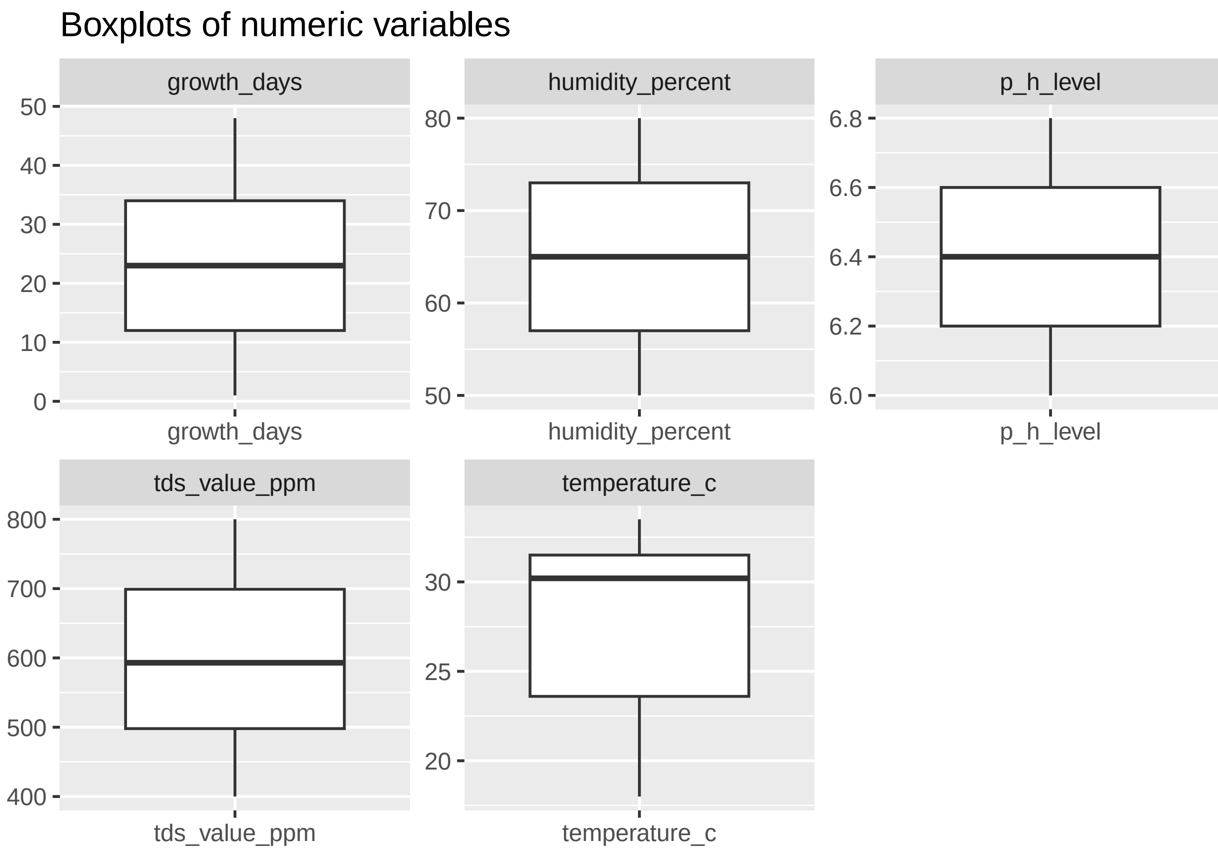
```
  ggplot(aes(x = value, fill = variable)) +
  geom_histogram(bins = 30, alpha = 0.6, color = "black", na.rm = TRUE) +
  facet_wrap(~ variable, scales = "free", ncol = 3) +
# theme_minimal() +
  guides(fill = "none")
```



Detect outliers plotting boxplots

```
df %>% select(where(is.numeric)) %>%
  pivot_longer(everything(),
               names_to = "variable",
               values_to = "value") %>%
  ggplot(aes(x = variable, y = value)) +
  geom_boxplot(outliers.alpha = 0.35) +
  labs(x = NULL, y = NULL, title = "Boxplots of numeric variables") +
  facet_wrap(~ variable, scales = "free", ncol = 3) +
  # theme_minimal() +
  guides(fill = "none")
```

```
## Warning in geom_boxplot(outliers.alpha = 0.35): Ignoring unknown parameters:
## `outliers.alpha`
```



Interpretation of the 2d stage EDA: Univariate analysis **Purpose**

This step examined each variable independently to describe its distribution, variation, and potential outliers.

Summary Statistics

Rows: 3,169

Columns: 7

Variable types: 1 Date (date), 1 Factor (plant_id), 5 Numeric (temperature_c, humidity_percent, tds_value_ppm, p_h_level, growth_days).

Missing data: 0 across all columns.

Date Variable

Range: 2023-08-03 → 2023-09-19 (48 unique days). → Represents about 7 weeks of continuous monitoring.

Plant ID

70 unique plants, each measured several times (roughly 45 observations per plant). → Confirms the dataset's panel/time-series nature.

Numeric Variables

N	Variable	Interpretation
1	temperature_c	Moderate variation — typical indoor growing temperature, stable across time.
2	humidity_percent	Reasonably humid environment — variation likely due to daily cycles.

N	Variable	Interpretation
3	tds_value_ppm	Stable nutrient concentration — healthy hydroponic range.
4	p_h_level	Excellent — near-neutral, stable pH.
5	growth_days	Data spans about 1.5 months: from 2023-08-03 to 2023-09-19.

Insights

No missing data or extreme anomalies.

Boxplots showed relatively symmetric distributions for most environmental variables, with slight skewness or mild outliers in TDS and Growth Days.

Overall, variability within variables supports robust analysis of growth-environment interactions.

Interpretation

Each variable behaves consistently with expectations for greenhouse data:

Environmental factors are well-controlled but still varied enough to analyze correlations with plant growth.

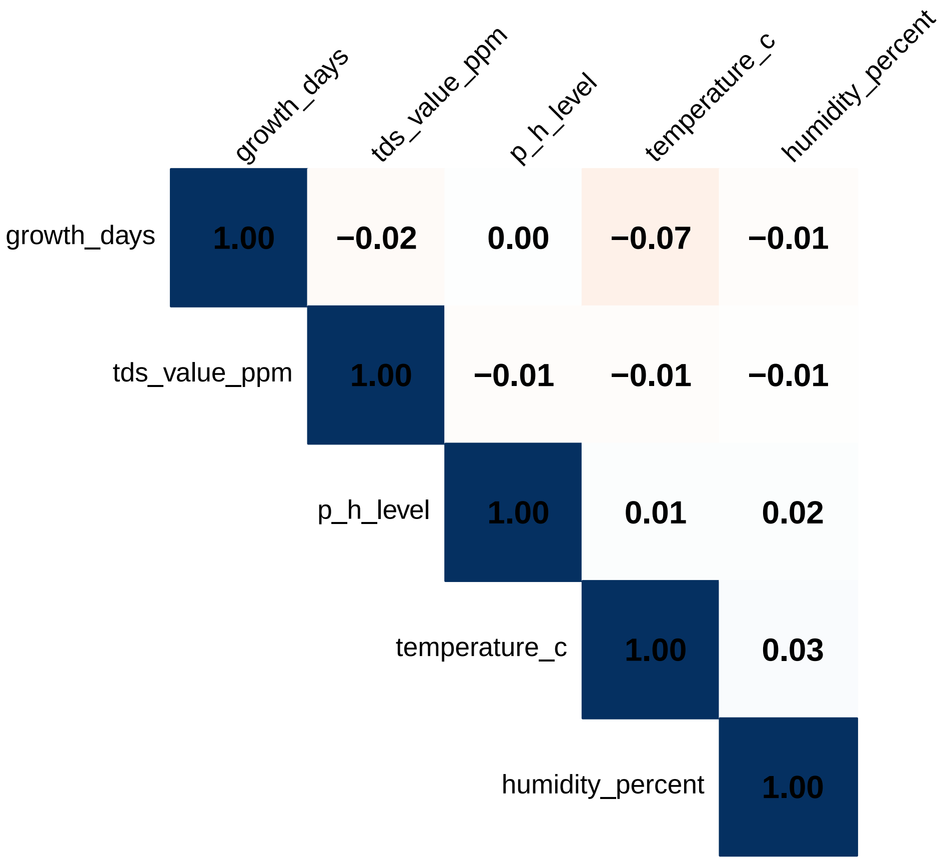
The dataset is statistically rich and suitable for later modeling (e.g., regression or correlation analysis).

Stage 3 of EDA: Bivariate/Multivariate analysis ("Understanding relationships" phase)

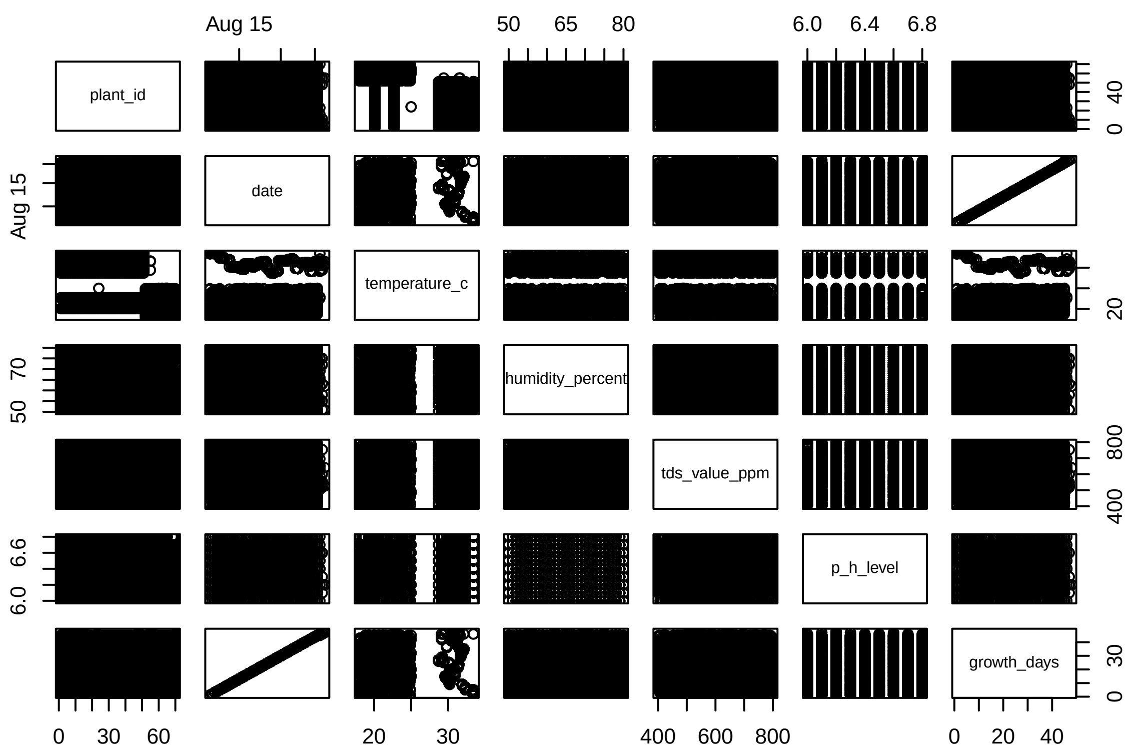
```
df %>% select(where(is.numeric)) %>%
  cor(use = "complete.obs") %>% round(2)
```

```
##                 temperature_c humidity_percent tds_value_ppm p_h_level
## temperature_c            1.00             0.03         -0.01      0.01
## humidity_percent         0.03             1.00         -0.01      0.02
## tds_value_ppm           -0.01            -0.01          1.00     -0.01
## p_h_level                0.01             0.02         -0.01      1.00
## growth_days             -0.07            -0.01         -0.02      0.00
##                 growth_days
## temperature_c         -0.07
## humidity_percent      -0.01
## tds_value_ppm         -0.02
## p_h_level              0.00
## growth_days            1.00
```

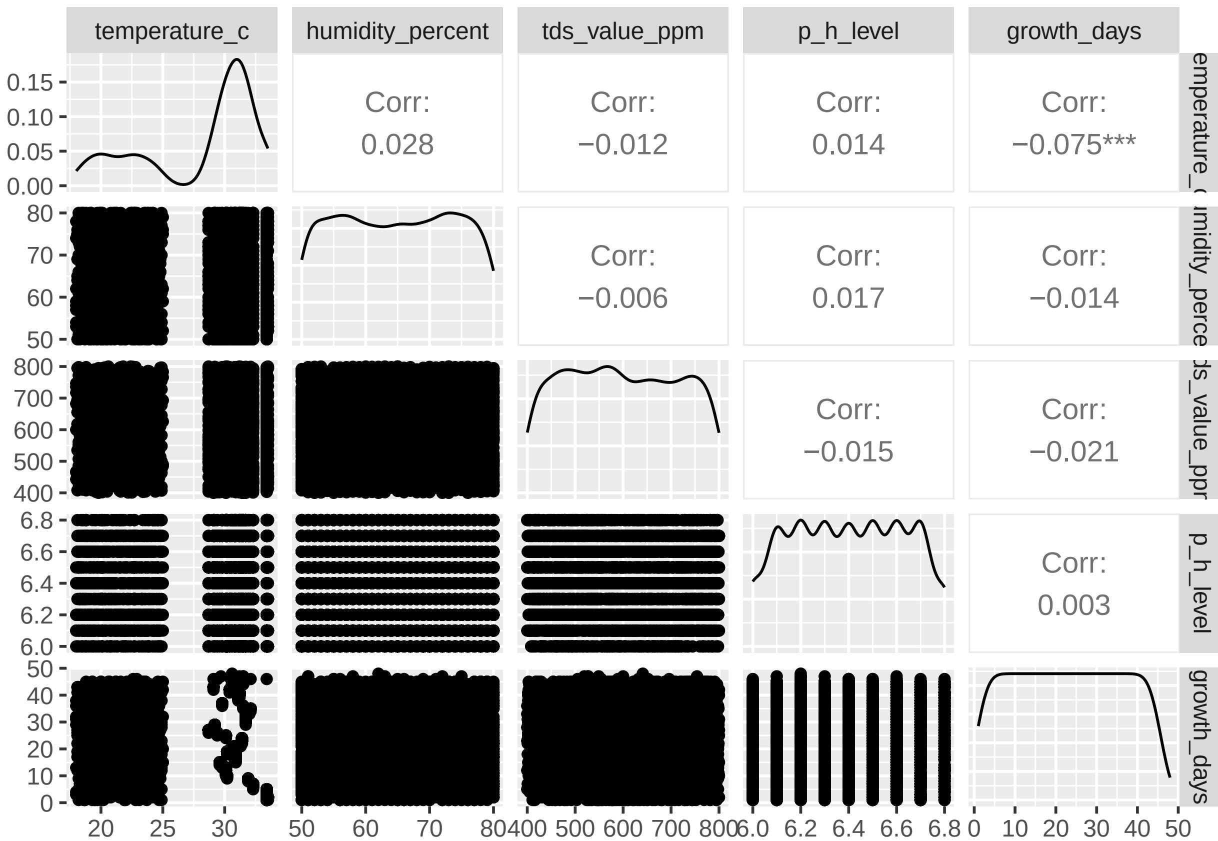
```
df %>% select(where(is.numeric)) %>%
  cor(use = "complete.obs") %>%
  corrplot::corrplot(method = "color",
                     type = "upper",
                     order = "hclust",
                     addCoef.col = "black",
                     tl.cex = 0.8,
                     tl.col = "black",
                     tl.srt = 45,
                     cl.pos = "n")
```


growth_days
tds_value_ppm
p_h_level
temperature_c
humidity_percent
growth_days
1.00
−0.02
0.00
−0.07
−0.01
tds_value_ppm
1.00
−0.01
−0.01
−0.01
p_h_level
1.00
0.01
0.02
temperature_c
1.00
0.03
humidity_percent
1.00

```r
pairs(df)
```


plant_id
date
temperature_c
humidity_percent
tds_value_ppm
p_h_level
growth_days
Aug 15
50 65 80
6.0 6.4 6.8
0 40
Aug 15
20
50 70
400 800
6.0 6.6
0 30
0 30 60
20 30
400 600 800
0 20 40

```r
df %>% select(-plant_id, -date) %>%
  ggpairs()
```



Purpose

This stage examined how environmental variables relate to each other and to the target variable — Growth Days.

Correlation Matrix (Rounded to 2 decimals)

Variable	temperature_c	humidity_percent	tds_value_ppm	p_h_level	growth_days
temperature_c	1.00	0.03	-0.01	0.01	-0.07
humidity_percent	0.03	1.00	-0.01	0.02	-0.01
tds_value_ppm	-0.01	-0.01	1.00	-0.01	-0.02
p_h_level	0.01	0.02	-0.01	1.00	0.00
growth_days	-0.07	-0.01	-0.02	0.00	1.00

Interpretation of Correlations

All correlations are weak ($|r| < 0.1$) — no strong linear dependencies among variables.

Slight negative correlation between temperature_c and growth_days (-0.07) could indicate that higher temperatures may slightly accelerate early growth, reducing total growth duration.

Other environmental factors (humidity_percent, tds_value_ppm, p_h_level) show minimal direct relationships to growth time.

The correlation plot confirms overall independence of predictors, meaning multicollinearity is not a concern.

General Interpretation

The weak correlations suggest lettuce growth in this dataset is not linearly dominated by any single factor — growth may depend on nonlinear or combined environmental effects (e.g., optimal ranges rather than simple linear relationships).

These findings justify future modeling efforts (like regression or machine learning) to explore interactions or thresholds rather than direct proportional effects.

Summary of the Three Phases

Phase	Focus	Key Findings
3.1 Data Discovery & Profiling	Structure & cleanliness	Data was complete, consistent, and properly formatted.
3.2 Univariate Analysis	Individual variable behavior	Variables showed natural variation without missing data; distributions were realistic.
3.3 Bivariate/Multivariate Analysis	Variable relationships	No strong linear relationships; temperature showed a slight negative association with growth days.

All statistics by plant id

```
all_stats_by_id <- df %>% group_by(plant_id) %>%
  summarise(
            date_min = min(date),
            date_mean = mean(date),
            date_median = median(date),
            date_max = max(date),
            temperature_c_min = min(temperature_c),
            temperature_c_q1 = quantile(temperature_c, 0.25),
            temperature_c_mean = mean(temperature_c),
            temperature_c_median = median(temperature_c),
            temperature_c_sd = sd(temperature_c),
            temperature_c_q3 = quantile(temperature_c, 0.75),
            temperature_c_max = max(temperature_c),
            humidity_percent_min = min(humidity_percent),
            humidity_percent_q1 = quantile(humidity_percent, 0.25),
            humidity_percent_mean = mean(humidity_percent),
            humidity_percent_median = median(humidity_percent),
            humidity_percent_sd = sd(humidity_percent),
            humidity_percent_q3 = quantile(humidity_percent, 0.75),
            humidity_percent_max = max(humidity_percent),
            tds_value_ppm_min = min(tds_value_ppm),
            tds_value_ppm_q1 = quantile(tds_value_ppm, 0.25),
            tds_value_ppm_mean = mean(tds_value_ppm),
            tds_value_ppm_median = median(tds_value_ppm),
            tds_value_ppm_sd = sd(tds_value_ppm),
            tds_value_ppm_q3 = quantile(tds_value_ppm, 0.75),
            tds_value_ppm_max = max(tds_value_ppm),
            p_h_level_min = min(p_h_level),
            p_h_level_q1 = quantile(p_h_level, 0.25),
            p_h_level_mean = mean(p_h_level),
            p_h_level_median = median(p_h_level),
            p_h_level_sd = sd(p_h_level),
            p_h_level_q3 = quantile(p_h_level, 0.75),
```

```r
            p_h_level_max = max(p_h_level),
            growth_days_min = min(growth_days),
            growth_days_q1 = quantile(growth_days, 0.25),
            growth_days_mean = mean(growth_days),
            growth_days_median = median(growth_days),
            growth_days_sd = sd(growth_days),
            growth_days_q3 = quantile(growth_days, 0.75),
            growth_days_max = max(growth_days)
  ) %>%
  mutate(
    across(where(is.numeric), ~ round(.x, 2))
         )

all_stats_by_id
```

```
## # A tibble: 70 x 40
##    plant_id date_min   date_mean  date_median date_max   temperature_c_min
##    <fct>    <date>     <date>     <date>      <date>                 <dbl>
##  1 1        2023-08-03 2023-08-25 2023-08-25  2023-09-16              20.1
##  2 2        2023-08-03 2023-08-25 2023-08-25  2023-09-16              20.1
##  3 3        2023-08-03 2023-08-26 2023-08-26  2023-09-18              20.1
##  4 4        2023-08-03 2023-08-26 2023-08-26  2023-09-19              20.1
##  5 5        2023-08-03 2023-08-25 2023-08-25  2023-09-16              20.1
##  6 6        2023-08-03 2023-08-25 2023-08-25  2023-09-16              20.1
##  7 7        2023-08-03 2023-08-26 2023-08-26  2023-09-18              20.1
##  8 8        2023-08-03 2023-08-25 2023-08-25  2023-09-16              20.1
##  9 9        2023-08-03 2023-08-25 2023-08-25  2023-09-17              20.1
## 10 10       2023-08-03 2023-08-25 2023-08-25  2023-09-16              20.1
## # i 60 more rows
## # i 34 more variables: temperature_c_q1 <dbl>, temperature_c_mean <dbl>,
## #   temperature_c_median <dbl>, temperature_c_sd <dbl>, temperature_c_q3 <dbl>,
## #   temperature_c_max <dbl>, humidity_percent_min <dbl>,
## #   humidity_percent_q1 <dbl>, humidity_percent_mean <dbl>,
## #   humidity_percent_median <dbl>, humidity_percent_sd <dbl>,
## #   humidity_percent_q3 <dbl>, humidity_percent_max <dbl>, ...
```

Summary of all statistics by plant id

```r
all_stats_by_id %>% summary()
```

```
##     plant_id     date_min             date_mean            date_median
##  1      : 1   Min.   :2023-08-03   Min.   :2023-08-25   Min.   :2023-08-25
##  2      : 1   1st Qu.:2023-08-03   1st Qu.:2023-08-25   1st Qu.:2023-08-25
##  3      : 1   Median :2023-08-03   Median :2023-08-25   Median :2023-08-25
##  4      : 1   Mean   :2023-08-03   Mean   :2023-08-25   Mean   :2023-08-25
##  5      : 1   3rd Qu.:2023-08-03   3rd Qu.:2023-08-25   3rd Qu.:2023-08-25
##  6      : 1   Max.   :2023-08-03   Max.   :2023-08-26   Max.   :2023-08-26
##  (Other):64
##     date_max          temperature_c_min temperature_c_q1 temperature_c_mean
##  Min.   :2023-09-16   Min.   :18.00     Min.   :18.80    Min.   :21.04
##  1st Qu.:2023-09-16   1st Qu.:18.32     1st Qu.:20.25    1st Qu.:21.94
##  Median :2023-09-16   Median :20.10     Median :30.10    Median :30.63
##  Mean   :2023-09-16   Mean   :19.56     Mean   :27.27    Mean   :28.14
##  3rd Qu.:2023-09-16   3rd Qu.:20.10     3rd Qu.:30.10    3rd Qu.:30.63
##  Max.   :2023-09-19   Max.   :20.10     Max.   :30.10    Max.   :30.67
```

```
##
##  temperature_c_median temperature_c_sd temperature_c_q3 temperature_c_max
##  Min.   :20.75        Min.   :1.810    Min.   :22.2     Min.   :24.60
##  1st Qu.:22.23        1st Qu.:2.277    1st Qu.:23.8     1st Qu.:26.65
##  Median :30.90        Median :2.380    Median :31.7     Median :33.50
##  Mean   :28.34        Mean   :2.301    Mean   :29.4     Mean   :31.25
##  3rd Qu.:30.90        3rd Qu.:2.380    3rd Qu.:31.7     3rd Qu.:33.50
##  Max.   :31.10        Max.   :3.160    Max.   :31.7     Max.   :33.50
##
##  humidity_percent_min humidity_percent_q1 humidity_percent_mean
##  Min.   :50.00        Min.   :53.00       Min.   :62.62
##  1st Qu.:50.00        1st Qu.:56.00       1st Qu.:63.71
##  Median :50.00        Median :57.00       Median :64.82
##  Mean   :50.19        Mean   :57.39       Mean   :64.88
##  3rd Qu.:50.00        3rd Qu.:59.00       3rd Qu.:65.81
##  Max.   :51.00        Max.   :62.50       Max.   :68.68
##
##  humidity_percent_median humidity_percent_sd humidity_percent_q3
##  Min.   :61.00           Min.   : 7.050      Min.   :68.00
##  1st Qu.:63.00           1st Qu.: 8.533      1st Qu.:71.00
##  Median :65.00           Median : 9.060      Median :72.88
##  Mean   :64.89           Mean   : 8.959      Mean   :72.29
##  3rd Qu.:66.00           3rd Qu.: 9.287      3rd Qu.:74.00
##  Max.   :72.00           Max.   :10.290      Max.   :76.00
##
##  humidity_percent_max tds_value_ppm_min tds_value_ppm_q1 tds_value_ppm_mean
##  Min.   :77.00        Min.   :400       Min.   :451.0    Min.   :567.0
##  1st Qu.:79.00        1st Qu.:402       1st Qu.:486.2    1st Qu.:584.2
##  Median :80.00        Median :405       Median :501.0    Median :598.3
##  Mean   :79.61        Mean   :409       Mean   :502.7    Mean   :598.0
##  3rd Qu.:80.00        3rd Qu.:412       3rd Qu.:515.8    3rd Qu.:611.9
##  Max.   :80.00        Max.   :449       Max.   :576.0    Max.   :628.2
##
##  tds_value_ppm_median tds_value_ppm_sd tds_value_ppm_q3 tds_value_ppm_max
##  Min.   :544.0        Min.   : 95.55   Min.   :641.0    Min.   :765.0
##  1st Qu.:574.2        1st Qu.:109.97   1st Qu.:680.0    1st Qu.:789.0
##  Median :596.5        Median :115.61   Median :698.8    Median :795.0
##  Mean   :596.9        Mean   :115.56   Mean   :695.0    Mean   :791.9
##  3rd Qu.:613.0        3rd Qu.:120.48   3rd Qu.:714.2    3rd Qu.:798.0
##  Max.   :666.0        Max.   :138.25   Max.   :749.0    Max.   :800.0
##
##  p_h_level_min    p_h_level_q1   p_h_level_mean  p_h_level_median
##  Min.   :6.000   Min.   :6.100   Min.   :6.310   Min.   :6.300
##  1st Qu.:6.000   1st Qu.:6.200   1st Qu.:6.380   1st Qu.:6.400
##  Median :6.000   Median :6.200   Median :6.400   Median :6.400
##  Mean   :6.001   Mean   :6.207   Mean   :6.399   Mean   :6.398
##  3rd Qu.:6.000   3rd Qu.:6.200   3rd Qu.:6.420   3rd Qu.:6.400
##  Max.   :6.100   Max.   :6.400   Max.   :6.490   Max.   :6.500
##
##   p_h_level_sd     p_h_level_q3   p_h_level_max   growth_days_min
##  Min.   :0.1900   Min.   :6.500   Min.   :6.700   Min.   :1
##  1st Qu.:0.2200   1st Qu.:6.600   1st Qu.:6.800   1st Qu.:1
##  Median :0.2300   Median :6.600   Median :6.800   Median :1
##  Mean   :0.2344   Mean   :6.598   Mean   :6.789   Mean   :1
```

```
##  3rd Qu.:0.2500   3rd Qu.:6.600   3rd Qu.:6.800   3rd Qu.:1
##  Max.   :0.2800   Max.   :6.700   Max.   :6.800   Max.   :1
##
##  growth_days_q1  growth_days_mean growth_days_median growth_days_sd
##  Min.   :12.00   Min.   :23.00    Min.   :23.00      Min.   :13.13
##  1st Qu.:12.00   1st Qu.:23.00    1st Qu.:23.00      1st Qu.:13.13
##  Median :12.00   Median :23.00    Median :23.00      Median :13.13
##  Mean   :12.07   Mean   :23.13    Mean   :23.14      Mean   :13.21
##  3rd Qu.:12.00   3rd Qu.:23.00    3rd Qu.:23.00      3rd Qu.:13.13
##  Max.   :12.75   Max.   :24.50    Max.   :24.50      Max.   :14.00
##
##  growth_days_q3  growth_days_max
##  Min.   :34.00   Min.   :45.00
##  1st Qu.:34.00   1st Qu.:45.00
##  Median :34.00   Median :45.00
##  Mean   :34.20   Mean   :45.26
##  3rd Qu.:34.00   3rd Qu.:45.00
##  Max.   :36.25   Max.   :48.00
##
```

Statistics (basic) by plant id

```
all_stats_by_id %>%
  select(plant_id,
         temperature_c_mean,
         humidity_percent_mean,
         tds_value_ppm_mean,
         p_h_level_mean,
         growth_days_max)
```

```
## # A tibble: 70 x 6
##    plant_id temperature_c_mean humidity_percent_mean tds_value_ppm_mean
##    <fct>                 <dbl>                 <dbl>              <dbl>
##  1 1                      30.6                  64.0               615.
##  2 2                      30.6                  64.6               597.
##  3 3                      30.6                  68.7               620.
##  4 4                      30.6                  63.4               598.
##  5 5                      30.6                  65.1               577.
##  6 6                      30.6                  64.4               614.
##  7 7                      30.6                  64.1               608.
##  8 8                      30.6                  66.3               598.
##  9 9                      30.6                  66.9               608.
## 10 10                     30.6                  65.8               613.
## # i 60 more rows
## # i 2 more variables: p_h_level_mean <dbl>, growth_days_max <dbl>
```

Statistics by maximum growth day

```
df %>% group_by(plant_id) %>%
  mutate(growth_days_max = max(growth_days)) %>%
  group_by(growth_days_max) %>%
  summarise(
    plant_id_n = n_distinct(plant_id),
            temperature_c_mean = mean(temperature_c),
            humidity_percent_mean = mean(humidity_percent),
            tds_value_ppm_mean = mean(tds_value_ppm),
```

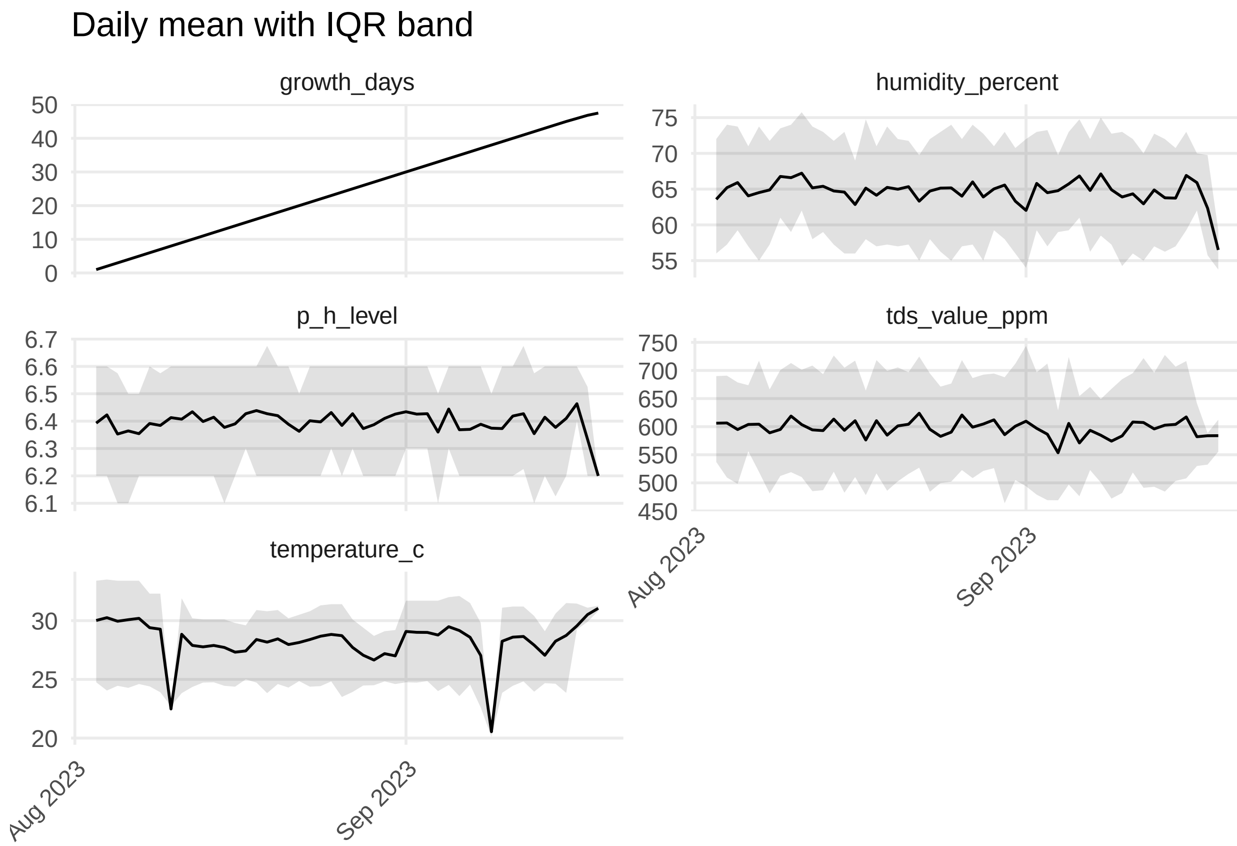
```
            p_h_level_mean = mean(p_h_level)
  ) %>%
  mutate(
    across(
      where(is.numeric), ~ round(.x, 2)
      )
  )
```

```
## # A tibble: 4 x 6
##   growth_days_max plant_id_n temperature_c_mean humidity_percent_mean
##             <dbl>      <dbl>              <dbl>                 <dbl>
## 1              45         59               28.1                  64.9
## 2              46          5               27.0                  65.0
## 3              47          5               28.8                  65.0
## 4              48          1               30.6                  63.4
## # i 2 more variables: tds_value_ppm_mean <dbl>, p_h_level_mean <dbl>
```

3.4 Visualize relationships

Time-series plots (trends over time)

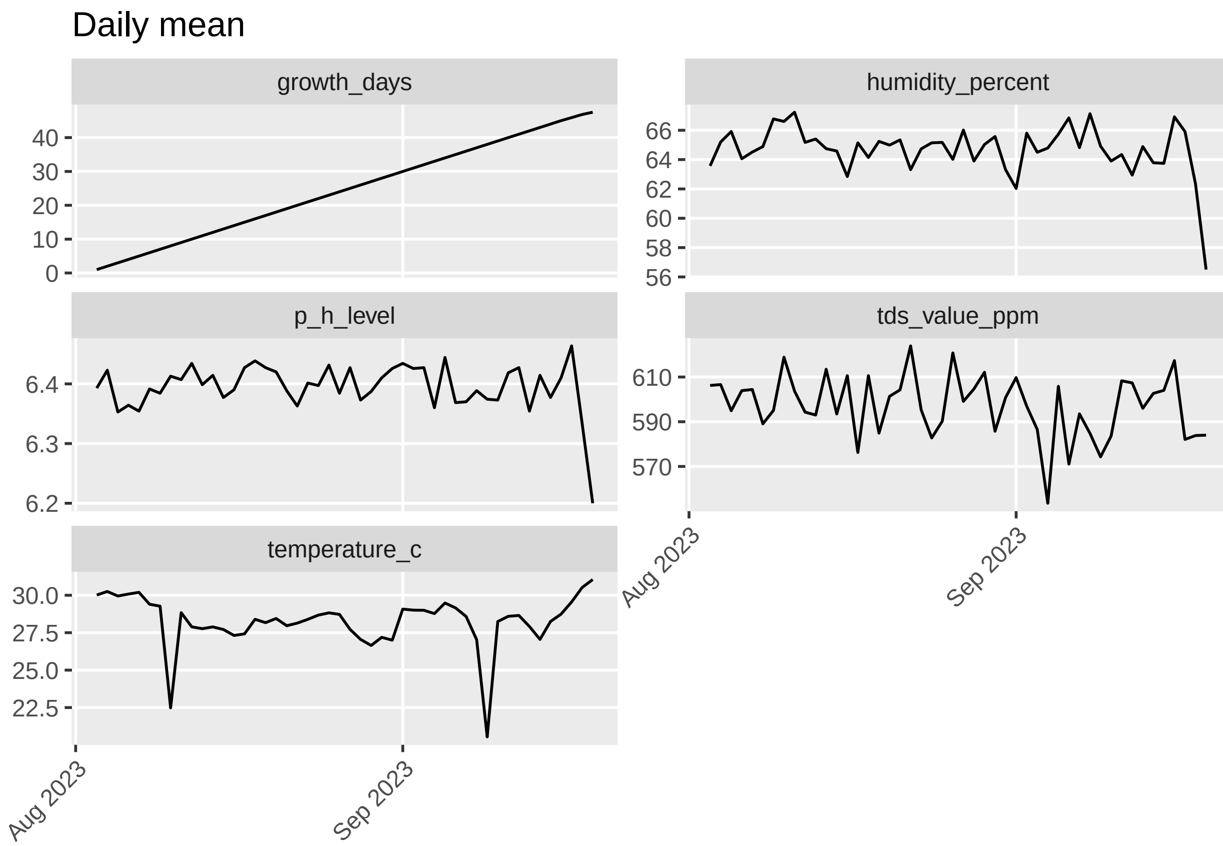
```
df %>%
  pivot_longer(-c(date, plant_id), names_to = "variable", values_to = "value") %>%
  group_by(variable, date) %>%
  summarise(
    mean = mean(value, na.rm = TRUE),
    p25  = quantile(value, 0.25, na.rm = TRUE),
    p75  = quantile(value, 0.75, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  arrange(variable, date) %>%
  ggplot(aes(date, mean)) +
  geom_ribbon(aes(ymin = p25, ymax = p75), alpha = 0.15) +
  geom_line() +
  facet_wrap(~ variable, scales = "free_y", ncol = 2) +
  scale_x_date(date_breaks = "1 month", date_labels = "%b %Y") +
  labs(x = NULL, y = NULL, title = "Daily mean with IQR band") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1),
        panel.grid.minor = element_blank())
```



c

```
df %>%
  pivot_longer(-c(date, plant_id), names_to = "variable", values_to = "values") %>%
  group_by(variable, date) %>%
  summarise(mean = mean(values)) %>%
  arrange(variable, date) %>%
  ggplot(aes(x = date, y = mean)) +
  geom_line() +
  facet_wrap(~ variable, scales = "free_y", ncol = 2) +
  scale_x_date(date_breaks = "1 month", date_labels = "%b %Y") +
  labs(x = NULL, y = NULL, title = "Daily mean") +
  # theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1),
        panel.grid.minor = element_blank())
```

```
## `summarise()` has grouped output by 'variable'. You can override using the
## `.groups` argument.
```

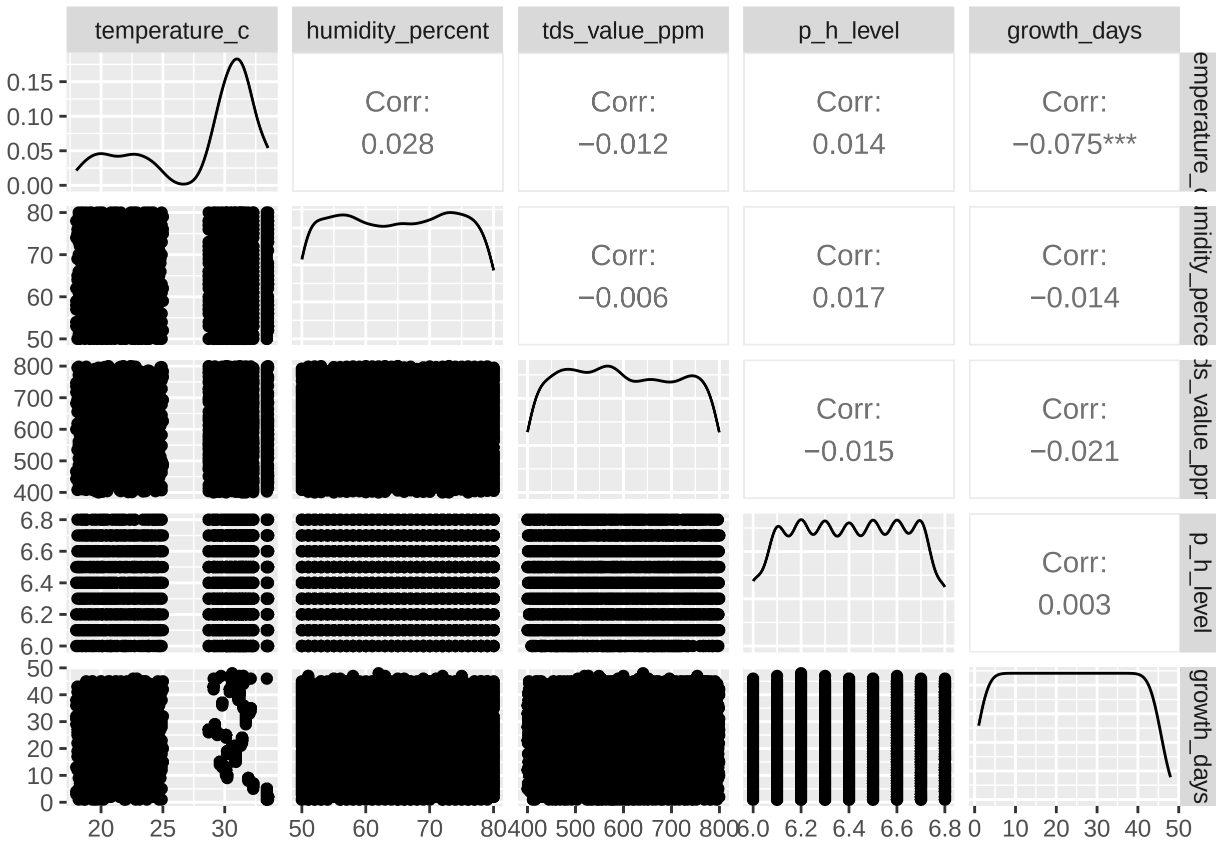


Scatterplots (check correlation)

```
df %>% select(where(is.numeric)) %>%
  ggpairs(progress = FALSE,
          upper = list(continuous = wrap("cor", method = "pearson")),
          diag = list(continuous = "densityDiag"),
          lwer = list(continuous = wrap("points", alpha = 0.4, size = 0.7))

  )
```

```
## Warning: The `...` argument of `ggpais()` is deprecated as of GGally 2.3.0.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



3.5 Form hypotheses

Does higher humidity improve growth?

Are weekly growth patterns consistent?

Is there seasonality in the data?